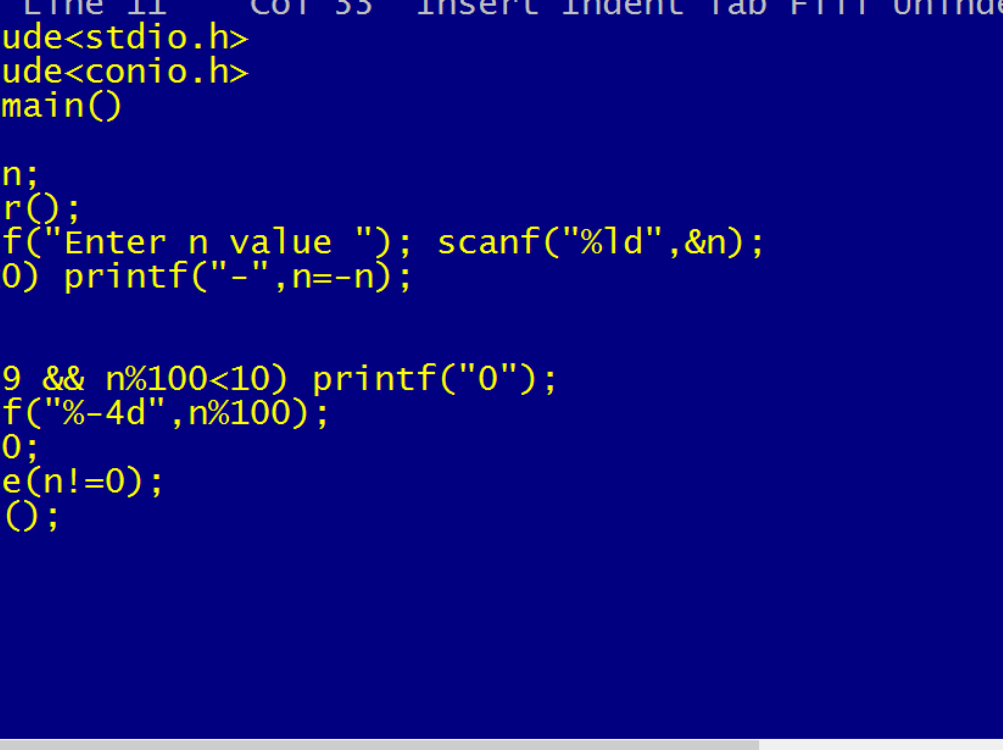
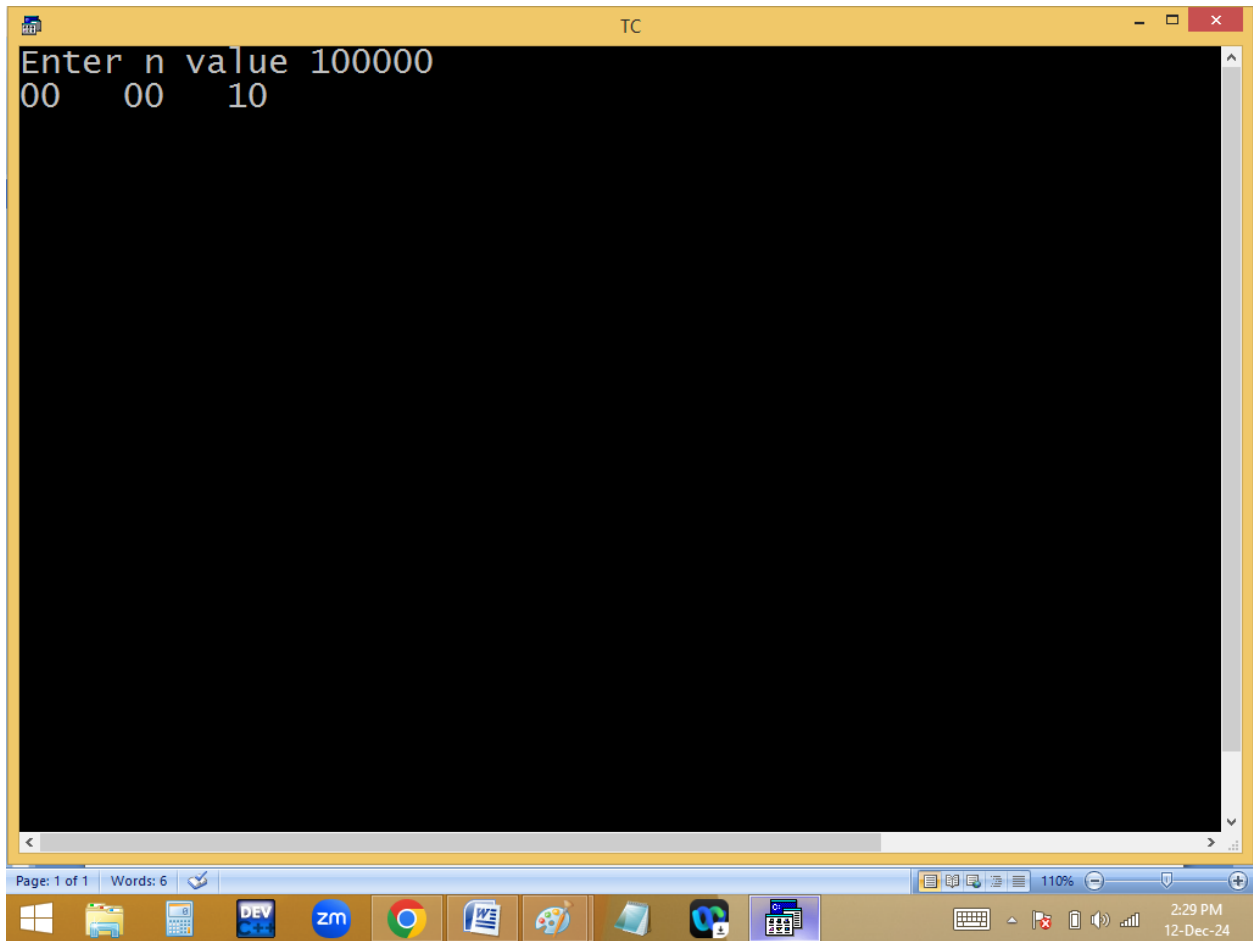


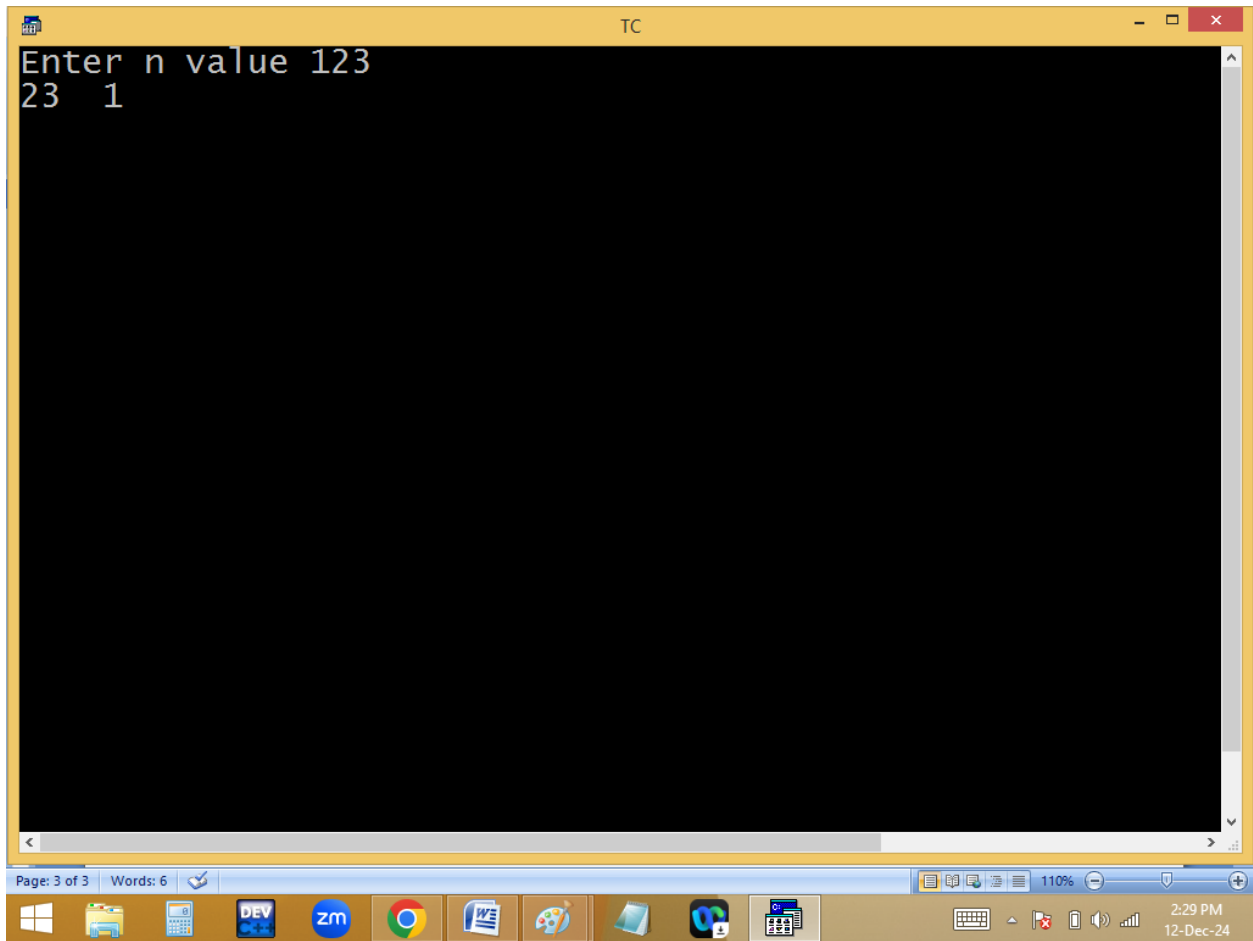
$$12345678 = 78\ 56\ 34\ 12$$


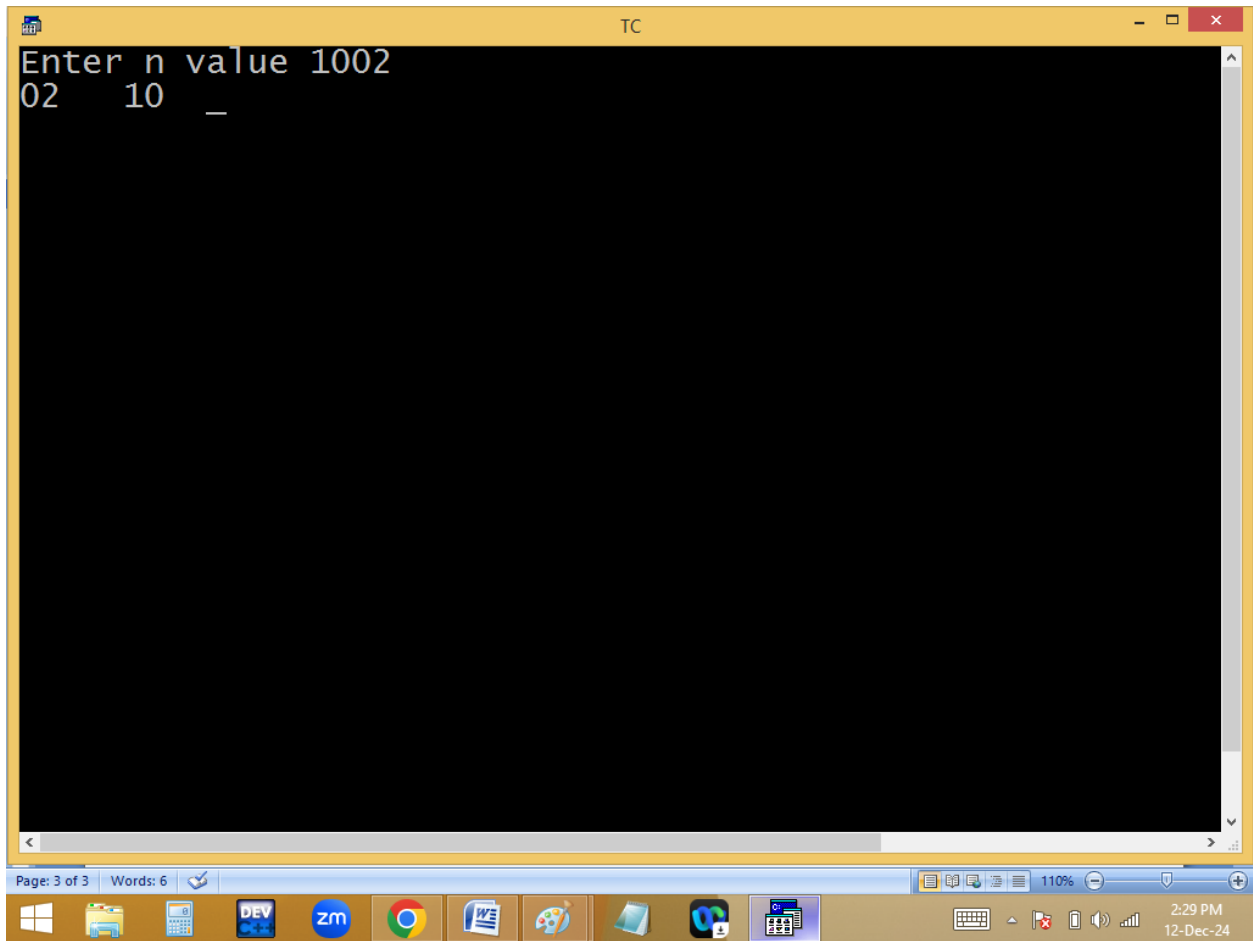
The screenshot shows the Turbo C++ IDE with a C program that prints the last two digits of a number. The code is as follows:

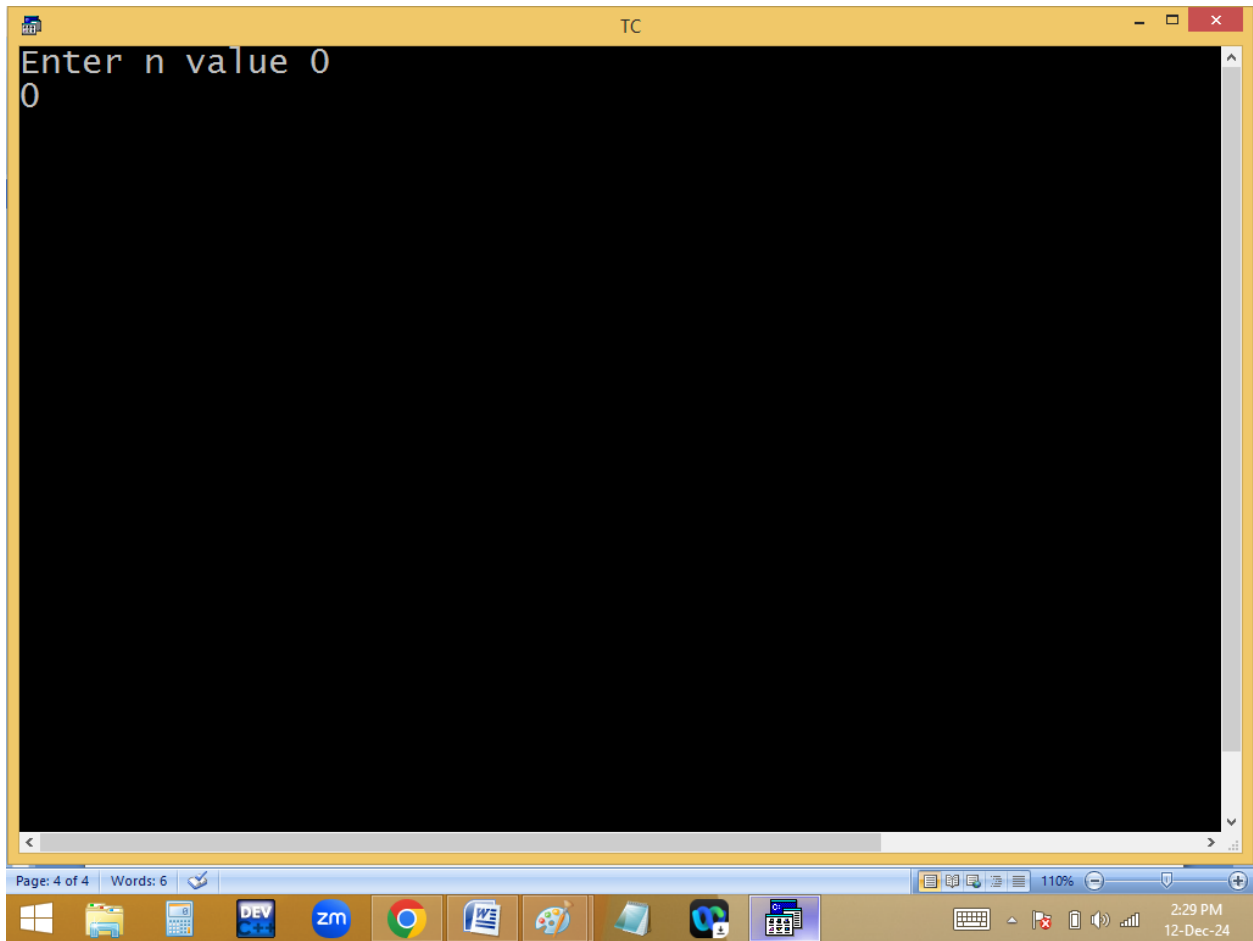
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    if(n<0) printf("-",n=-n);
    do
    {
        if(n>9 && n%100<10) printf("0");
        printf("%-4d",n%100);
        n/=100;
    }while(n!=0);
    getch();
}
```

The IDE interface includes a menu bar with File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom indicates the window size is 1894 x 763px and the zoom level is 100%.









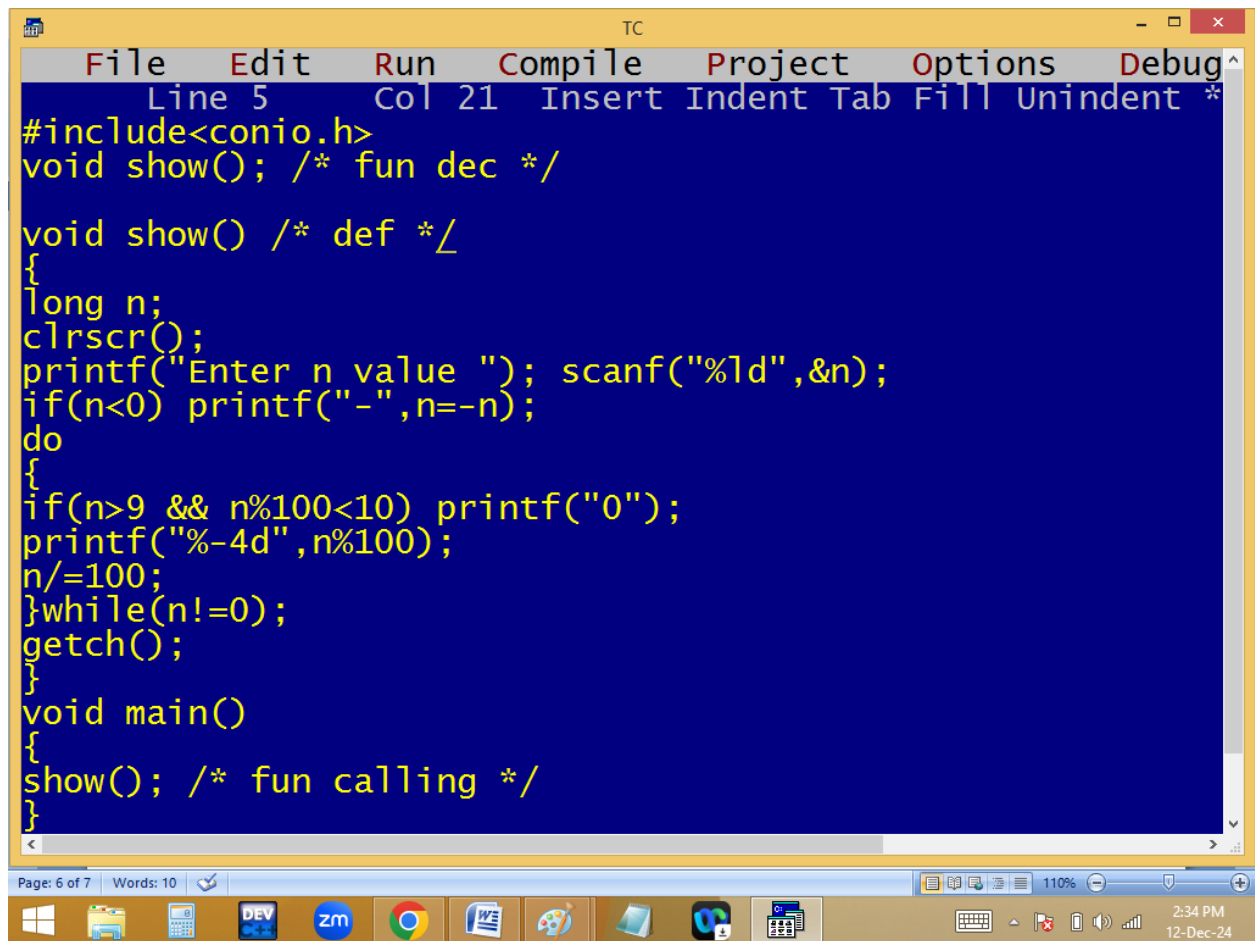
```
TC
Enter n value -1200
-00 12
```

Page: 5 of 5 Words: 6

2:29 PM 12-Dec-24

✓ ✓ ✗ ✓ $1234 \% 100 = 34$
if($n > 9$ && $n \% 100 < 10$) p(0); $1234 / 100 = 12$
~~100~~ $\% 100 = 00$

$123 \% 100 = 23$
 $102 \% 100 = 02$
9 ✓



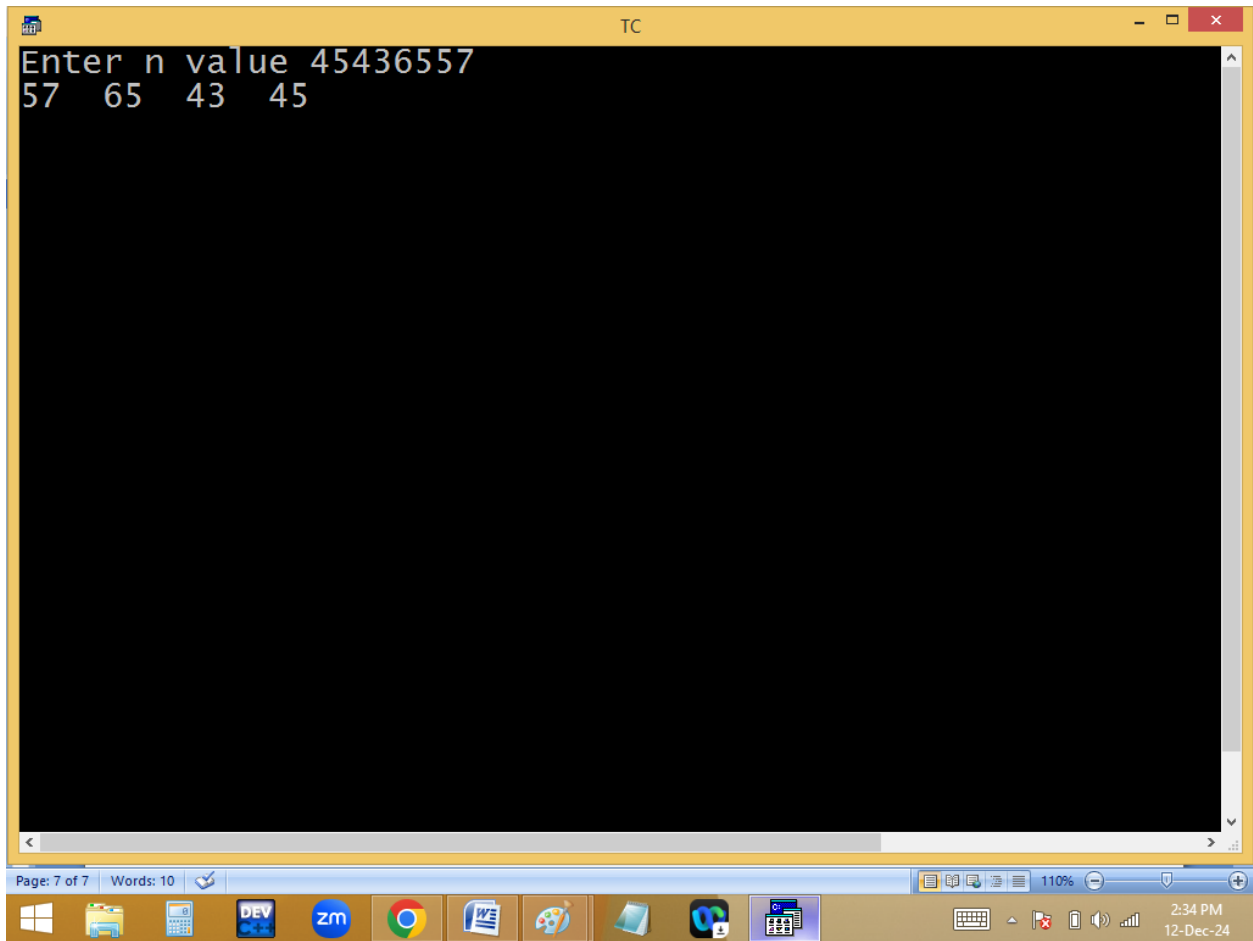
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a menu bar with 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', and 'Debug'. Below the menu bar, the status bar indicates 'Line 5', 'Col 21', and various editing options like 'Insert', 'Indent', 'Tab', 'Fill', and 'Unindent'. The main editing area has a dark blue background with yellow text. The code is as follows:

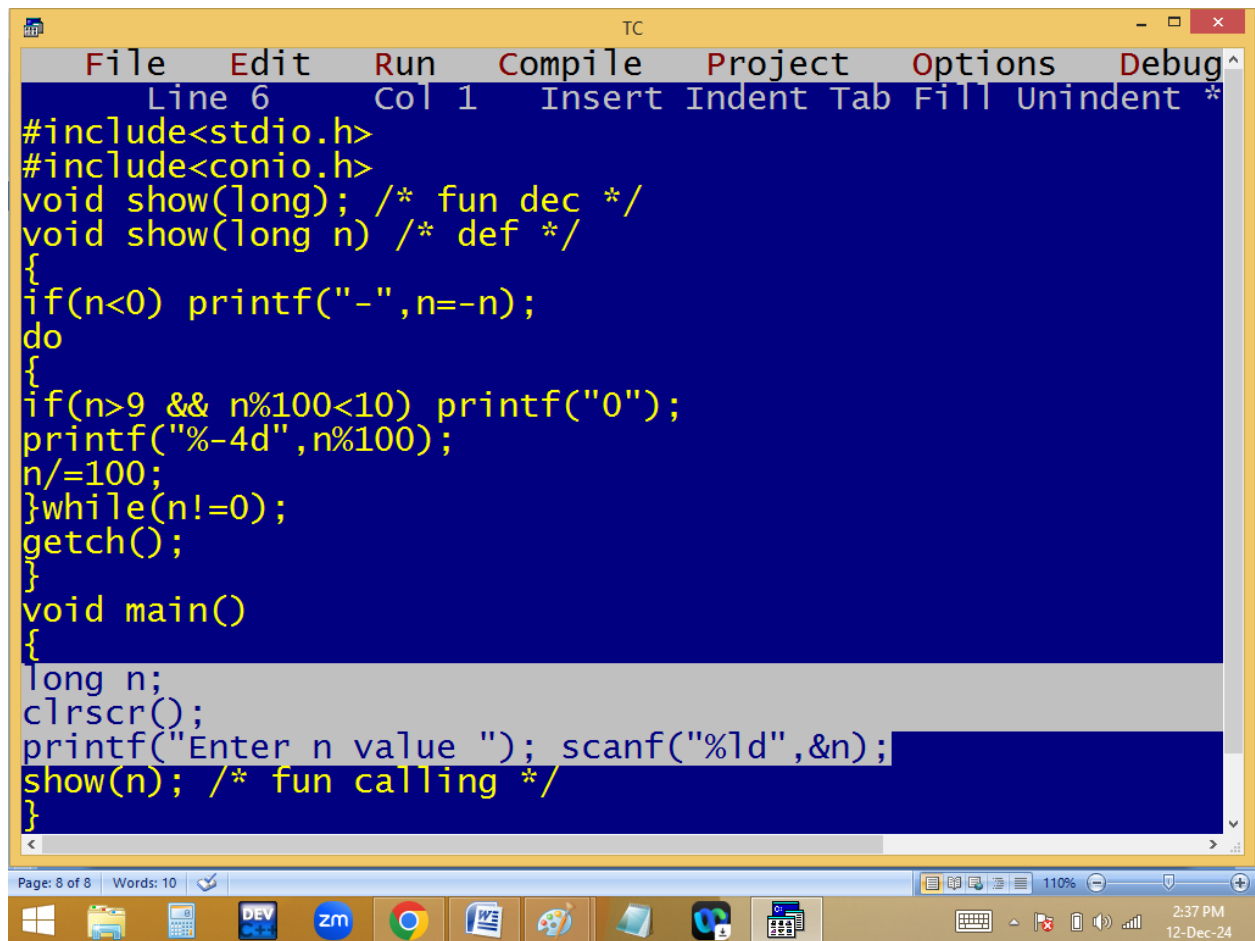
```
#include<conio.h>
void show(); /* fun dec */

void show() /* def */
{
    long n;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    if(n<0) printf("-",n=-n);
    do
    {
        if(n>9 && n%100<10) printf("0");
        printf("%-4d",n%100);
        n/=100;
    }while(n!=0);
    getch();
}

void main()
{
    show(); /* fun calling */
}
```

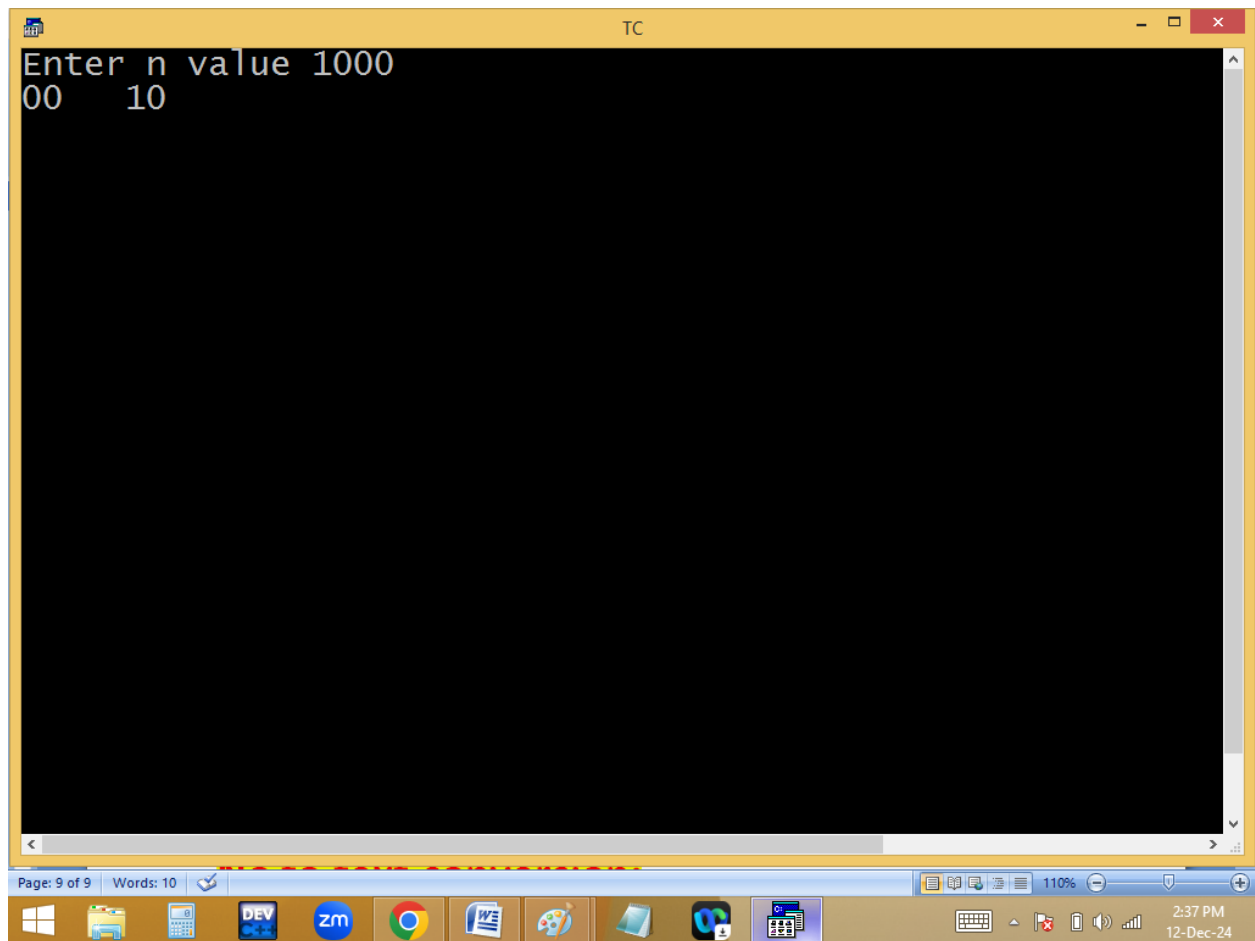
At the bottom of the window, there is a status bar showing 'Page: 6 of 7' and 'Words: 10'. Below the status bar is a Windows taskbar with various icons including Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner shows '2:34 PM' and '12-Dec-24'.





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 1", and "Insert Indent Tab Fill Unindent *". The code is written in C and is designed to reverse a number. It includes headers for `stdio.h` and `conio.h`. A function `show` is declared and then defined. The function takes a `long` integer `n` and prints it with a leading dash if it's negative, then prints it in reverse order using `printf` with a format of `0` for leading zeros and `%-4d` for the digits. It uses a `do-while` loop to process the digits. The `main` function declares a `long` variable `n`, clears the screen with `clrscr()`, prompts the user to enter a value, and calls the `show` function. The bottom status bar shows "Page: 8 of 8" and "Words: 10". The Windows taskbar at the bottom includes icons for various applications and the system clock showing 2:37 PM on 12-Dec-24.

```
#include<stdio.h>
#include<conio.h>
void show(long); /* fun dec */
void show(long n) /* def */
{
if(n<0) printf("-",n=-n);
do
{
if(n>9 && n%100<10) printf("0");
printf("%-4d",n%100);
n/=100;
}while(n!=0);
getch();
}
void main()
{
long n;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
show(n); /* fun calling */
}
```



No to text conversion:

#include<stdio.h>

#include<conio.h>

void main()

{

long n,m,rev=0;

```
clrscr();

printf("Enter a no "); scanf("%ld",&n);

if(n<0)printf("-",n=-n); m=n;

while(m!=0){int
r=m%10;rev=rev*10+r;m=m/10;} /*rev
no*/

do

{

switch(rev%10)

{

case 0: printf("Zero");break;

case 1: printf("One");break;

case 2: printf("Two");break;

case 3: printf("Three");break;
```

```
case 4: printf("Four");break;
case 5: printf("Five");break;
case 6: printf("Six");break;
case 7: printf("Seven");break;
case 8: printf("Eight");break;
case 9: printf("Nine");break;
}

printf(" "); rev=rev/10;

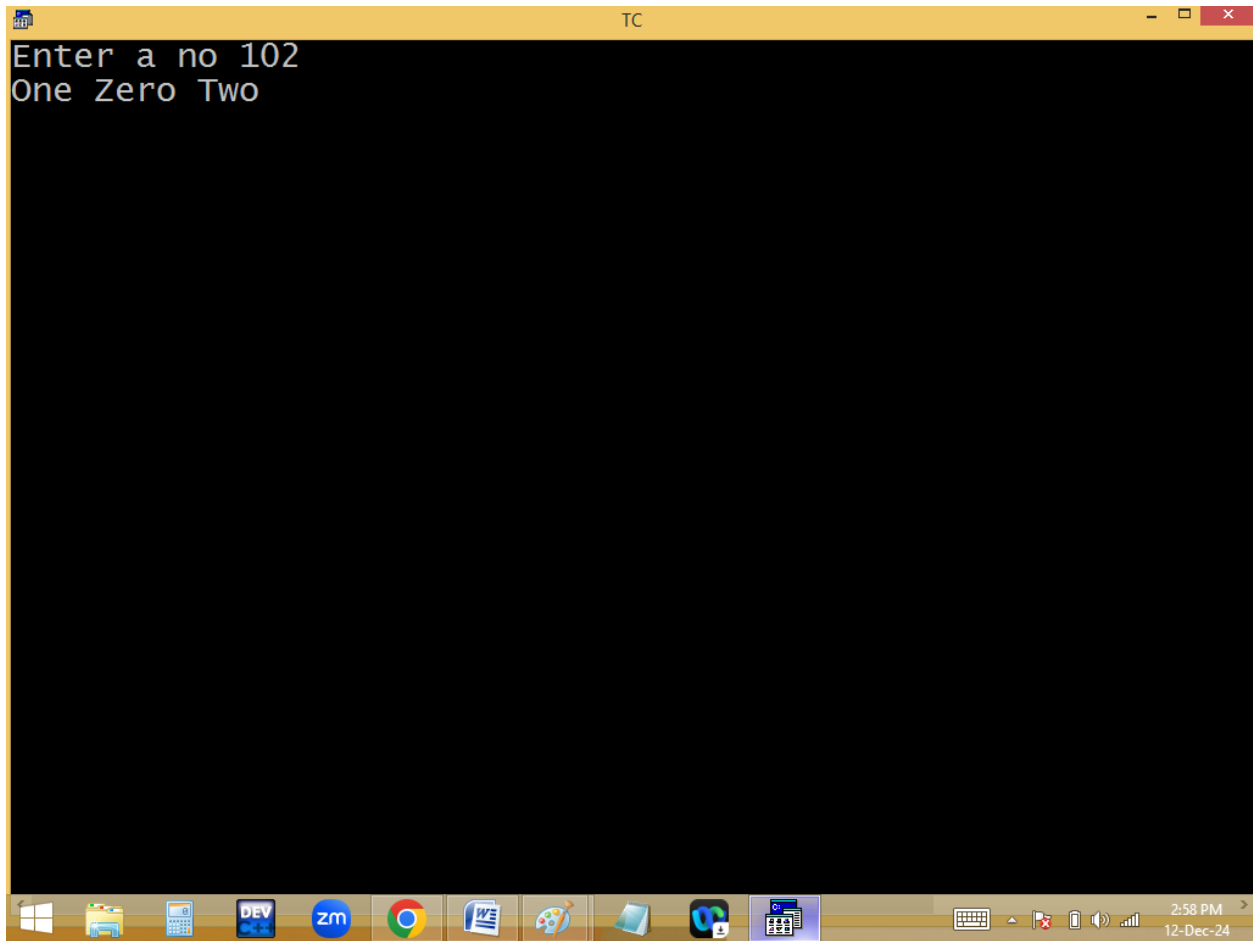
}while(rev!=0);

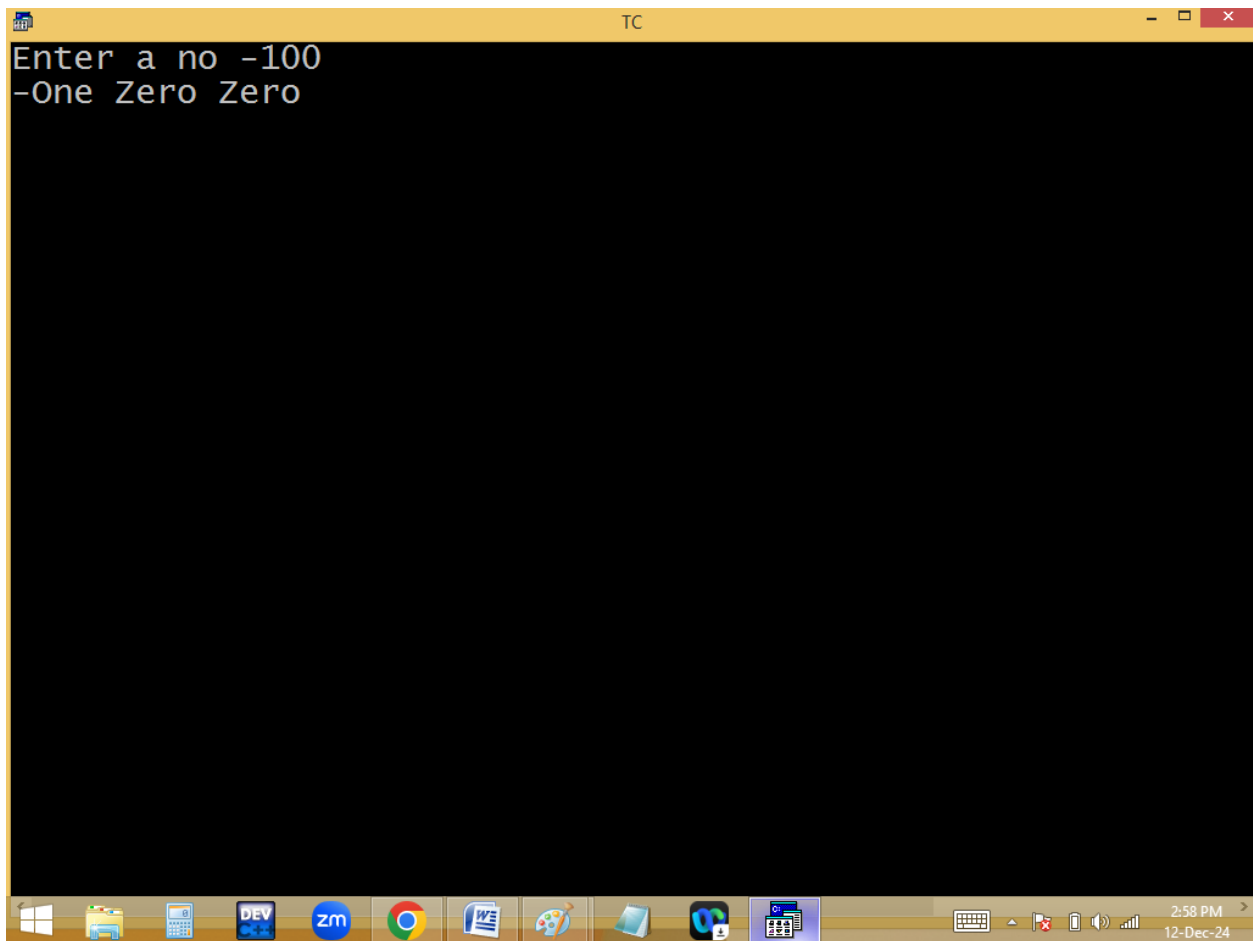
while(n!=0 && n%10==0)printf("Zero
",n=n/10);

getch();

}
```







```

do
{
switch(rev%10)
{
case 0: p(zero);break;
case 1: p(one);b; ←
case 2: p(two);b;
case 9: p(nine);
}
printf(" ");
rev=rev/10;
}while(rev!=0);

```

$$\begin{array}{r} n \\ 102 \\ \hline \end{array}$$

$$\begin{array}{r} rev \\ 2017 \div 10 = 1 \text{ one} \\ 20 \div 10 = 2 \text{ two} \\ 1 \div 10 = 0 \text{ zero} \end{array}$$

$$\begin{array}{r} n \\ 100 \div 10 = 0 \\ 10 \div 10 = 0 \\ 1 \div 10 = 1 \end{array}$$

one zero zero

while (n!=0 && n%10==0) p(zero , n/=10);

for loop:

It is an entry control loop.

for is a keyword.

It is also used to repeat a program several times based on a condition.

When compared with while and do while, for loop is looking to be smart. In for it is compulsory to maintain two semicolons.

For works without condition also and default condition is always 1 i.e. true.

Generally for loop is having 3 expressions.

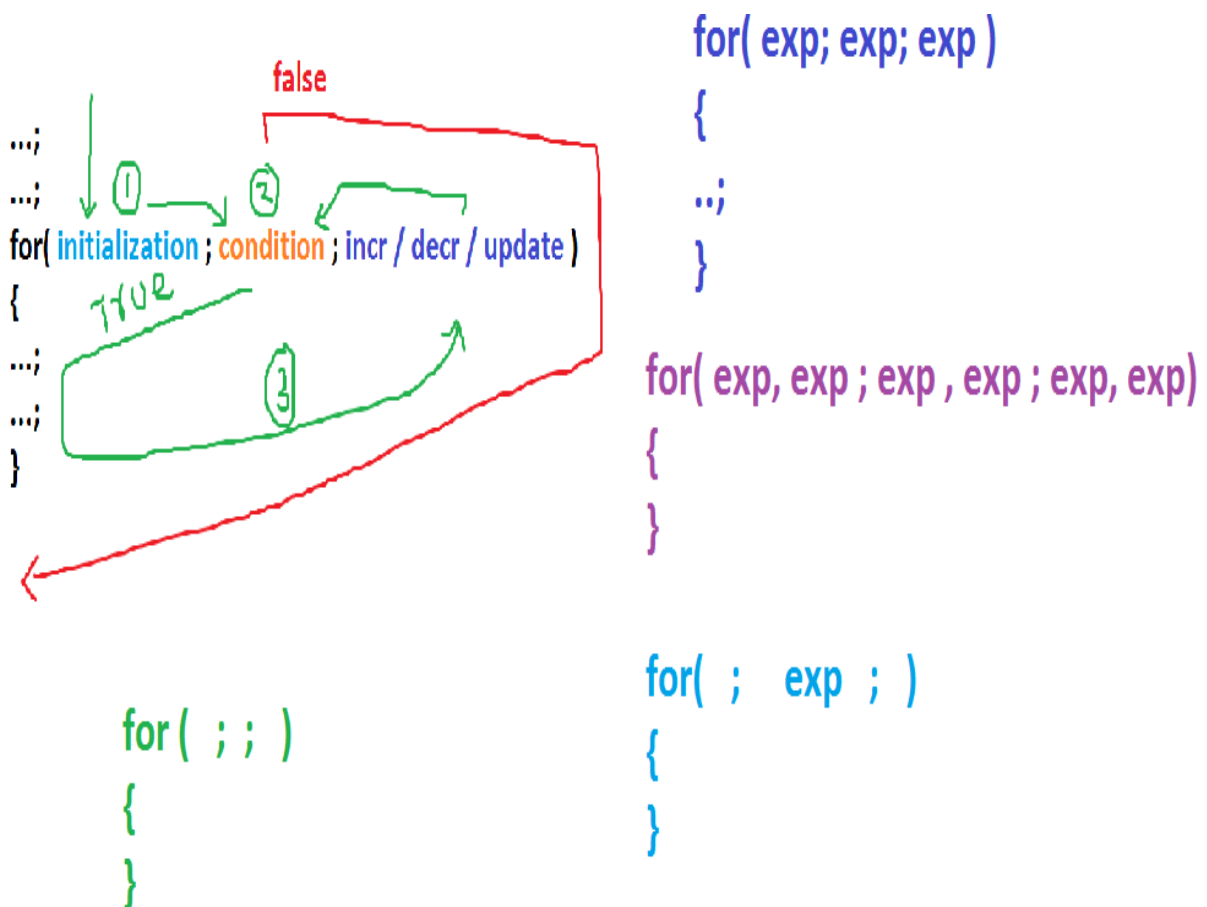
- 1. Initialization**
- 2. Test condition / expression**
- 3. Increment/decrement / updation**

At first entry of for loop the initialization part is executed and later the test condition is checked. If the condition is true then the for block statements are executed. After completion of the block, the increment or decrement part is executed. Later once again the test condition is evaluated. If it is true then once again for block statements are executed. Like this the process is continued until the condition becomes false. Here the initialization part is executed only once, at the time of loop beginning.

It is mandatory to maintain 2 semicolon (;) in a for loop.

If the for loop is having more than three expressions, it is mandatory to separate the expressions with , separator.

If the for loop is having less than three expressions, then leave the expressions with empty semicolon.



Printing the given table:

Example 9th table

$$9*1=9$$

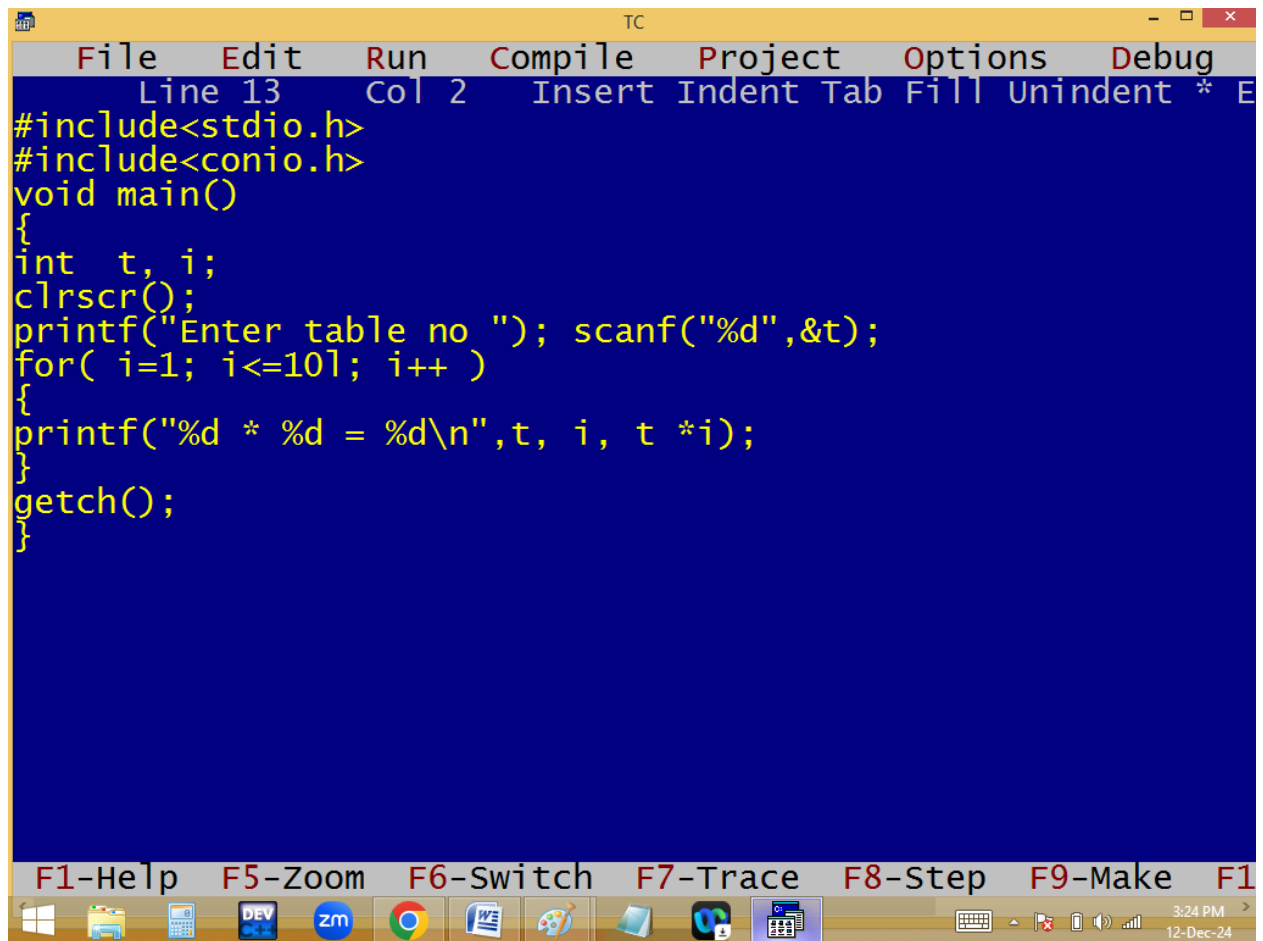
$$9*2=18$$

$$9*3=27$$

..

..

$$9*10=90$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 2", and "Insert Indent Tab Fill Unindent * E". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t, i;
    clrscr();
    printf("Enter table no "); scanf("%d",&t);
    for( i=1; i<=10; i++ )
    {
        printf("%d * %d = %d\n",t, i, t *i);
    }
    getch();
}
```

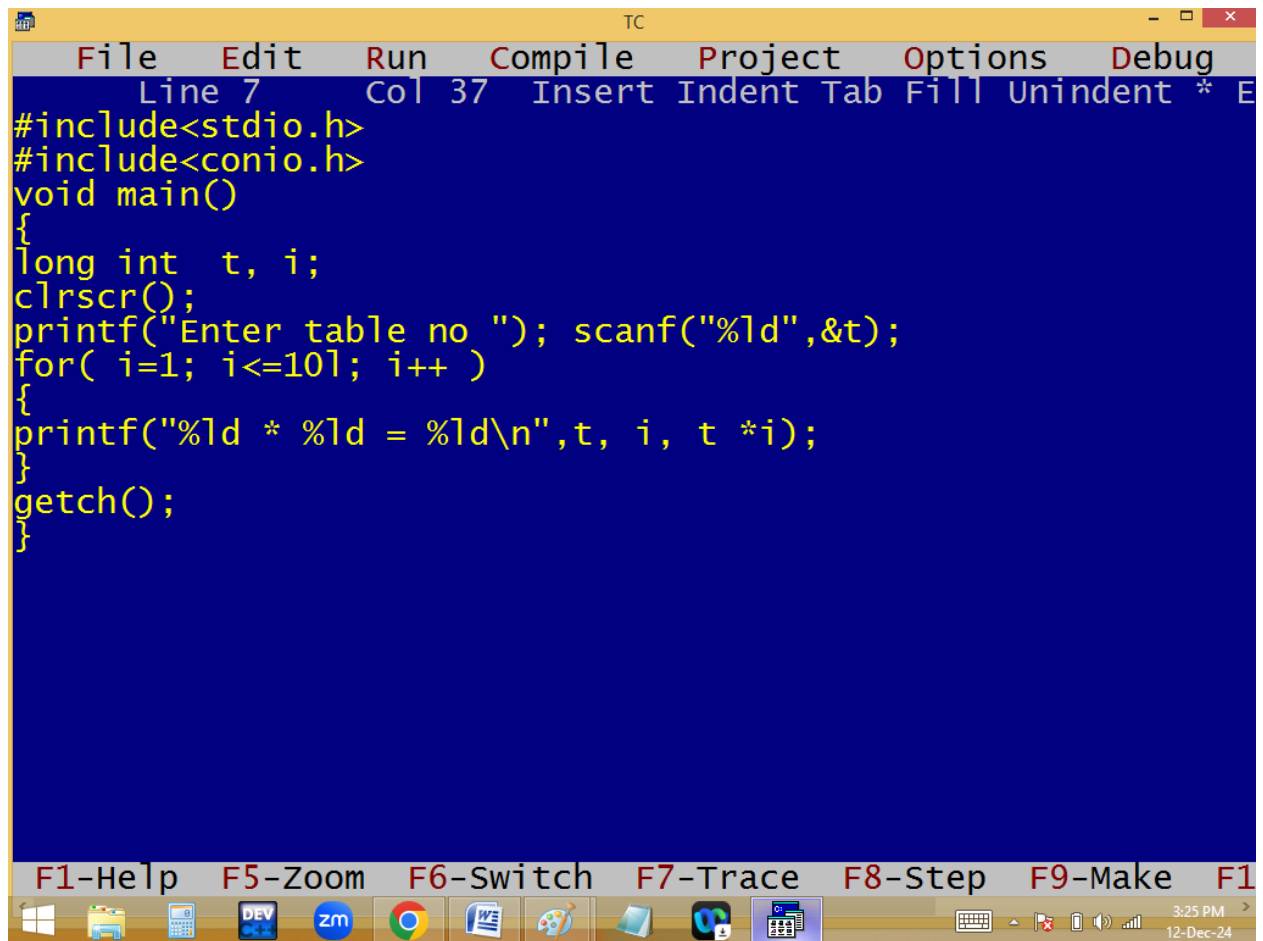
Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other background applications. The system tray on the right shows the time as "3:24 PM" and the date as "12-Dec-24".

```
TC
Enter table no 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```



```
TC
Enter table no 4000
4000 * 1 = 4000
4000 * 2 = 8000
4000 * 3 = 12000
4000 * 4 = 16000
4000 * 5 = 20000
4000 * 6 = 24000
4000 * 7 = 28000
4000 * 8 = 32000
4000 * 9 = -29536
4000 * 10 = -25536
```





The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 7", "Col 37", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is a C program that prompts the user to enter a table number and then prints the multiplication table for that number from 1 to 10. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int  t, i;
    clrscr();
    printf("Enter table no "); scanf("%ld",&t);
    for( i=1; i<=10; i++ )
    {
        printf("%ld * %ld = %ld\n",t, i, t * i);
    }
    getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing the Start button, several application icons (including File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, Paint, and the TC IDE), and the system clock displaying "3:25 PM" and "12-Dec-24".

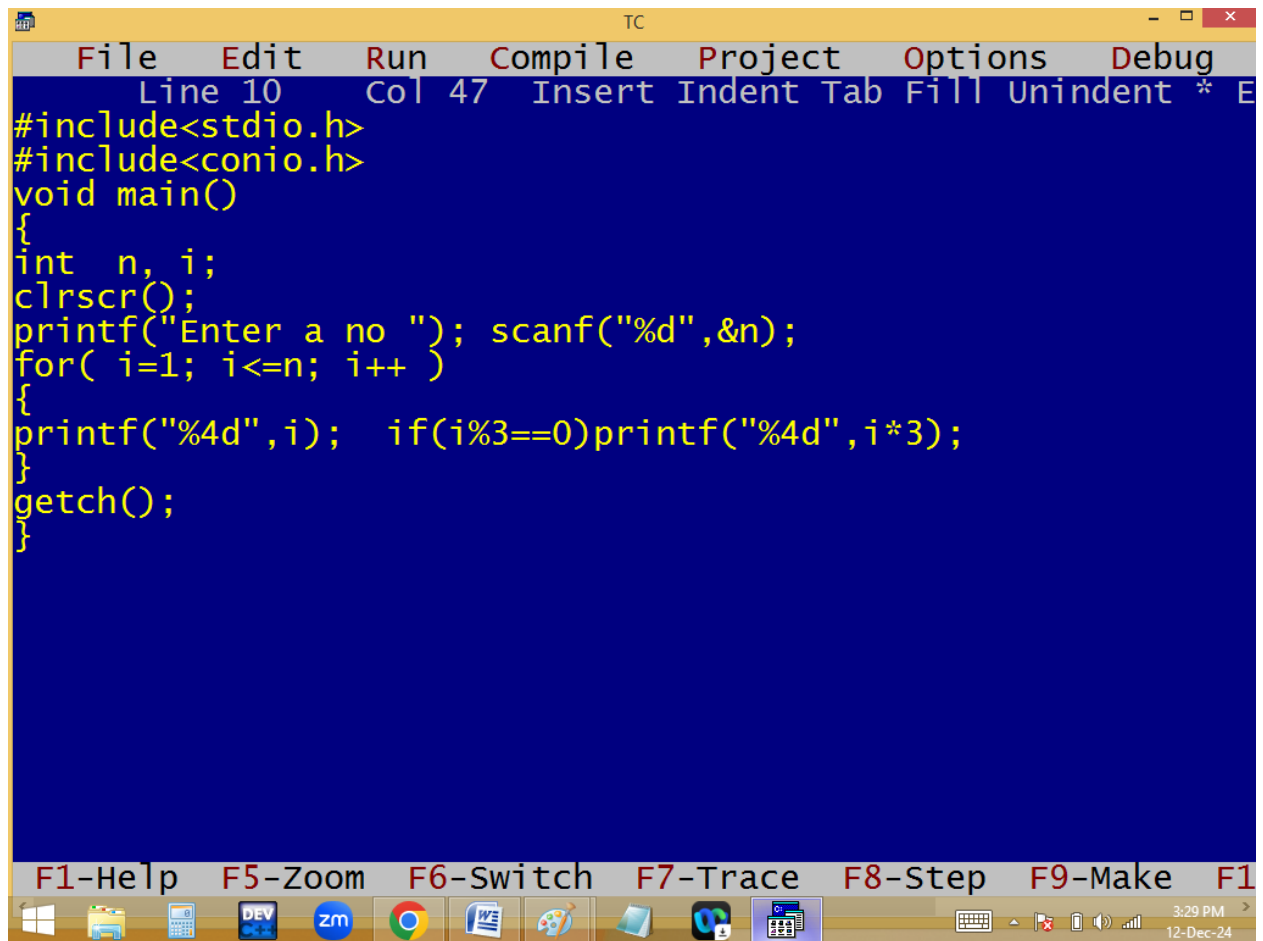
```
Enter table no 400000
400000 * 1 = 400000
400000 * 2 = 800000
400000 * 3 = 1200000
400000 * 4 = 1600000
400000 * 5 = 2000000
400000 * 6 = 2400000
400000 * 7 = 2800000
400000 * 8 = 3200000
400000 * 9 = 3600000
400000 * 10 = 4000000
```

$$\frac{t}{9} * \frac{i}{1} = \frac{t \times i}{9}$$
$$9 \times 1 = 9$$

```
for( i=1; i<=10; i++ )
{
    9 * 1 = 9
    printf("%d * %d = %d\n", t, i, t * i);
}
```

Printing below series:

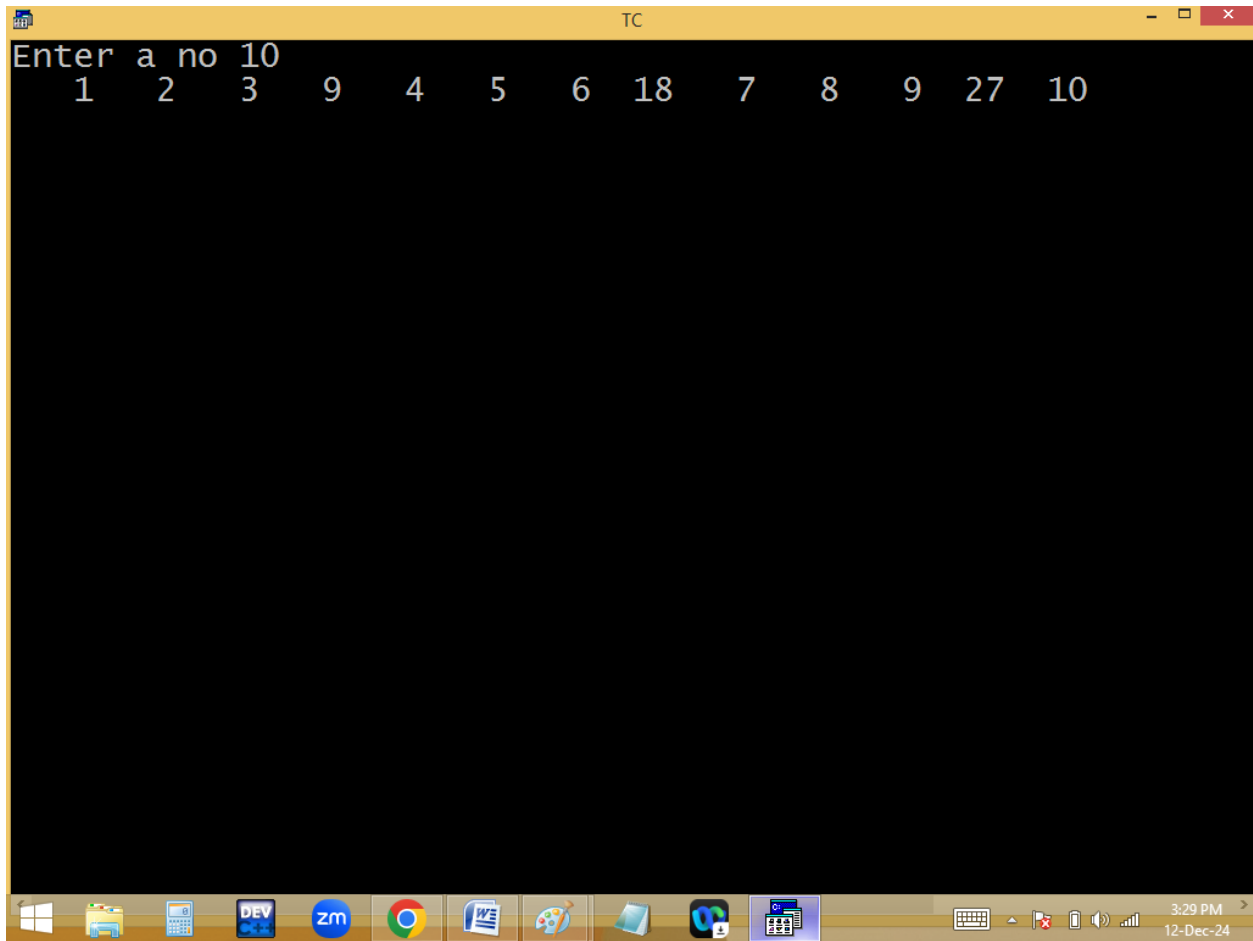
1 2 3 9 4 5 6 18 7 8 9 27 ... n



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 10", "Col 47", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n, i;
    clrscr();
    printf("Enter a no "); scanf("%d",&n);
    for( i=1; i<=n; i++ )
    {
        printf("%4d",i);  if(i%3==0)printf("%4d",i*3);
    }
    getch();
}
```

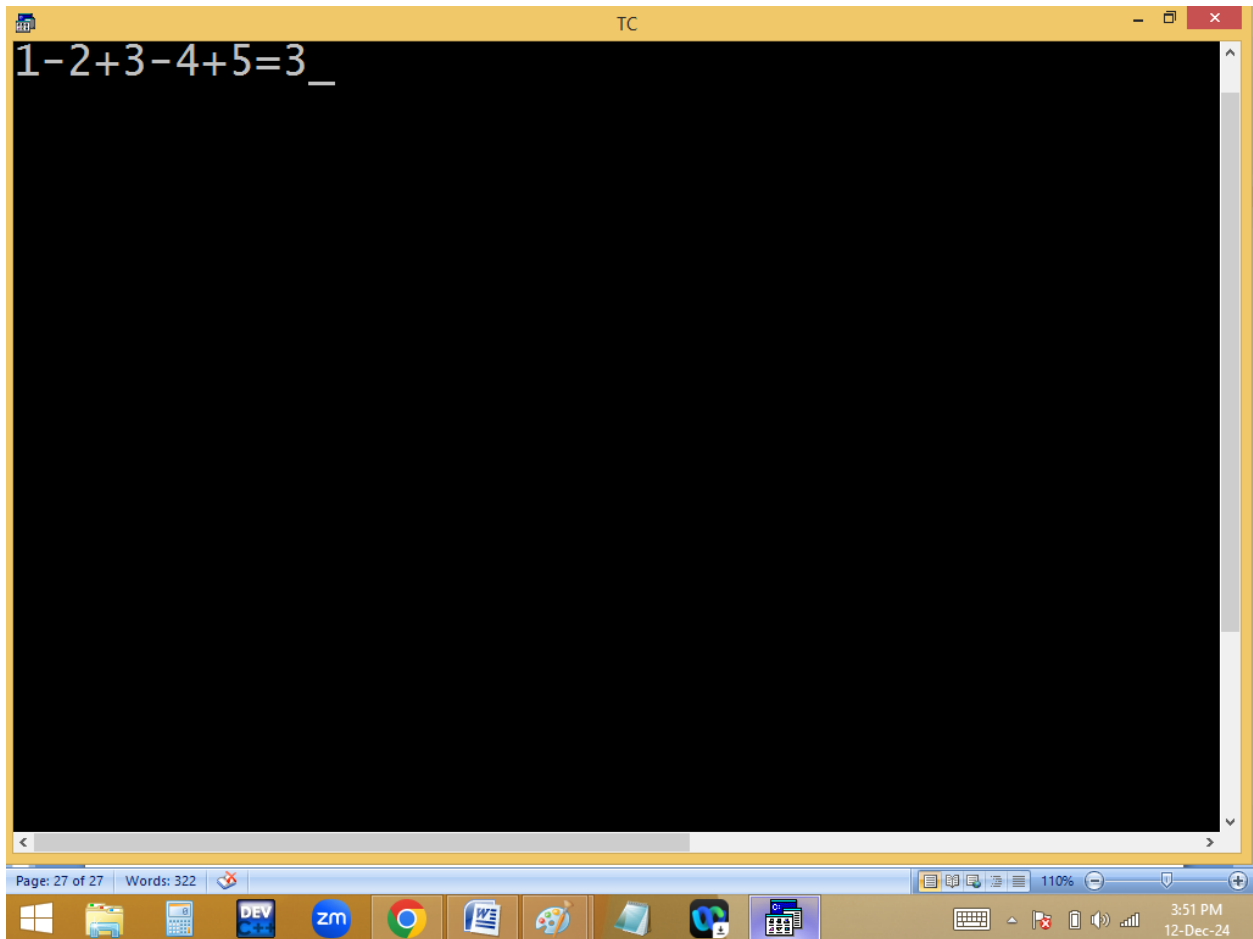
Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". At the very bottom is the Windows taskbar, which includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as "3:29 PM" and the date as "12-Dec-24".



```
TC
#include<stdio.h>
#include<conio.h>
void main()
{
int  n, i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for( i=1; i<=n; i++ )
{
if(i%2==0)printf("%d+",i,s-=i);
else printf("%d-",i,s+=i);
}
printf("\b=%d",s);
getch();
}
```

1894 x 763px 100%

3:51 PM 12-Dec-24



$$\begin{array}{r} 1+3+5 = 9 \\ -2+-4 = -6 \\ \hline 3 \end{array}$$

$$n=5 \implies 1-2+3-4+5=3$$

```

for(i=0;i<=n;i++)
{
✓ if( i%2==0 ) p("%d+",s+=i);
  else p("%d-",i,s+=i);
}
p("\b=%d",s);

```

<u>n</u>	<u>i</u>
5	1
	2
	3
	4
	5

<u>s</u>
0+1=1
1-2=-1
-1+3=2
+2-4=-2
-2+5=3

$$1-2+3-4+5 = 3$$

Home work:

1. $n = 5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

2. $n = 5 \rightarrow 1 + 1/1 + 1/2 + 1/3 + 1/4 + 1/5 = 3.28$